

Zaud Rehman

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SUMMARY

Software Engineer with hands-on experience in systems programming, backend development, and open-source contributions in Rust and Python. Delivered production-grade APIs, real-time WebSocket systems, and probabilistic data structure libraries through internships and personal projects, with a strong foundation in memory-safe concurrency and performance optimization.

EDUCATION

Srinivas University

B.Tech in Computer Science Engineering | CGPA: 9.08/10.0

Mangaluru, Karnataka, India

Dec 2021 – Apr 2025

Karnataka Secondary Education Examination Board (KSEEB)

12th Grade, Percentage: 80%

Karnataka, India

2020

TECHNICAL SKILLS

Languages: Rust, Python, Java, C, C++, JavaScript/TypeScript, SQL

Frameworks & Technologies: Axum, FastAPI, Django REST, Next.js, WebSocket, REST APIs, React

Databases & Caching: PostgreSQL, MongoDB, Redis, SQLx

Cloud & Tools: Docker, Git, Oracle Cloud (OCI), JWT Authentication, Microservices, Cargo

PROJECTS

BloomCraft | Rust | GitHub

Nov 2025 – Present

- Building a production-grade Bloom filter library implementing 12 variants (Standard, Scalable, Partitioned, Sharded, etc.) for high-throughput concurrent systems.
- Engineered a three-tier concurrency model using **AtomicU64** CAS for lock-free inserts and **RwLock** striping with Lemire's fast range reduction.
- Integrated pluggable hash strategies (WyHash, XXH3, SIMD), zero-copy binary serde, and HyperLogLog cardinality estimation via a type-state builder API.

NoteFlow | Rust, Axum, Next.js 14, PostgreSQL, Redis, WebSocket | Live | Backend

Aug – Dec 2025

- Built a full-stack collaborative note-taking platform with a Rust/Axum backend sustaining 1,000+ concurrent WebSocket connections at sub-100ms latency.
- Implemented JWT auth with refresh token rotation, Redis-based rate limiting, and a rich Markdown editor with LaTeX support achieving sub-second cross-device sync.

Aesthete | React 18, Three.js, Zustand, TailwindCSS | Live | GitHub

Jan 2026 – Present

- Developed a 3D algorithm visualization engine rendering 25+ algorithms (DP, Graphs, Sorting) at 60 FPS using WebGL and Three.js.
- Designed a generator-based **AlgorithmController** for time-travel debugging, frame-perfect stepping, and synchronized multi-language code highlighting.

INTERNSHIPS

Backend Development Intern

Tikanga Pvt. Limited

May 2024 – Jun 2024

Mangaluru, Karnataka, India

- Developed and maintained RESTful APIs for Tikanga's mobile application using Django REST Framework, covering user management, authentication, and business logic endpoints.
- Reduced bug rate by 15% through comprehensive unit and integration testing, systematic debugging, and code review participation.

Artificial Intelligence & Machine Learning Intern

Thaniya Technologies

Sep 2023

Mangaluru, Karnataka, India

- Built data preprocessing pipelines for 10,000+ records using Python and scikit-learn, including cleaning, feature engineering, and normalization.
- Achieved 10% improvement in model prediction accuracy through systematic hyperparameter tuning and cross-validation.

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified Data Science Professional – Oracle (2025)

Problem Solving (Intermediate) – HackerRank (2025)

AWS Artificial Intelligence Practitioner Learning – Amazon Web Services (2025)

Microsoft Azure AI Fundamentals (AI-900) – Certiport (2024)